Request for Information (RFI)

Title: Mission Integration, Launch, and Operations of GeoXO Satellites

This RFI is for informational purposes only. It is not a request for proposals or a solicitation for a contract or grant award, and it does not obligate the Government in any way. The Government will not reimburse respondents for any costs associated with responding to this request.

Introduction

The National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC), on behalf of the National Oceanic and Atmospheric Administration (NOAA) Geostationary Earth Orbit Observations (GEO) Program, is seeking information from qualified vendors capable of providing mission integration and operations support, including launch services, payload processing facility and services, mission operations, ground systems, and antenna services for GeoXO mission-critical weather satellite. This RFI is intended to gather market intelligence, technical approaches, and rough-order-of-magnitude (ROM) considerations to inform a potential future procurement.

Background

The National Oceanic and Atmospheric Administration (NOAA) operates a system of Geostationary Earth Orbit (GEO) satellites to provide continuous weather imagery and monitoring of meteorological data to protect life and property across the Western Hemisphere. Two NOAA Geostationary Earth Orbit (GEO) satellites remain operational at all times providing coverage for the eastern United States and most of the Atlantic Ocean and the western United States and Pacific Ocean basin. The Geostationary Extended Observations (GeoXO) Program is the follow-on to the Geostationary Operational Environmental Satellite (GOES)-R series. The GeoXO initial launch is planned for 2032. GeoXO is a collaborative development and acquisition effort between NOAA and NASA. Program oversight activities occur at National Environmental Satellite, Data, and Information Service (NESDIS) Headquarters and the NASA Goddard Space Flight Center (GSFC). The acquisition of the end-to-end GeoXO system includes spacecraft, instruments, launch services, mission operations, and all associated ground system elements. The NOAA GeoXO satellites will provide critical environmental observations for products supporting weather forecasting and warnings, and safe and efficient commercial and private air and marine transportation. Each GeoXO satellite will carry advanced Imager and Sounder instruments, and a Data Collection System (DCS) payload. The GeoXO satellite provides continuous, mission critical weather data in support of national and operational users. High availability, operational resilience, and rapid anomaly response are essential due to the mission's criticality.

Mission Integration Prime (MIP) Concept of Operations (CONOPS)

The NASA GeoXO Flight Project under the NOAA GEO Program will be procuring a "Mission Integration Prime" capability to manage and perform mission integration functions for GeoXO series satellites. These functions were performed and managed by the Government for previous GOES series. The MIP CONOPS below is one approach to performing the GeoXO mission integration services. The Government is analyzing alternative MIP approaches and will use the responses to this RFI to inform the analysis, identify new alternatives, and select the optimal MIP approach.

The Mission Integration Prime (MIP) contractor will provide the following:

Launch Vehicle Services (Fixed Price CLIN)

- Two launch vehicles to carry the GeoXO-1 and GeoXO-2 satellites
- Launch campaign planning, management, licensing, reviews, operations, and countdown execution
- Launch vehicle integration requirements development, verification, engineering, and analysis
- Satellite to launch vehicle integration

Payload Processing Services (Fixed Price CLIN)

- Payload processing facility to process the GeoXO-1 and GeoXO-2 satellites
- Payload processing facility management, planning, reviews, requirements development and verification
- Support the GeoXO spacecraft, launch vehicle, and instrument vendors in logistics, processing, testing, and spacecraft fueling at the payload processing facility

Mission Operations Services (Fixed-Price CLINs with Performance Incentives)

- Mission operations facilities, antennas, ground systems, and personnel to operate the GeoXO-1 and GeoXO-2 satellites from lift-off through 10 years of on-orbit operations for each satellite
- Pre-launch mission readiness and end-to-end space to ground compatibility testing
- Pre-launch operations product development and validation
- All satellite command and control of the GeoXO-1 and GeoXO-2 satellites from lift-off through 10 years of on-orbit operations
- Uplink and downlink from a Federal Communications Commission (FCC) L-band protected ground station
- Satellite operations from a Federal Information Security Management Act (FISMA) compliant ground system with a FIPS-200 categorization of "High" based on integrity and availability
- Real-time operations (nominal and contingency) from the MIP operations facilities
- Spacecraft and instrument activation and post-launch testing
- Instrument Level 1b (L1b) science product generation, where simplified product level descriptions are listed below
    - Level 0 (L0) data – Raw, unprocessed instrument and spacecraft Consultative Committee for Space Data Systems (CCSDS) packetized science data needed to generate L1b data
    - Level 1b (L1b) data – Radiometrically calibrated and geo-located (navigated) instrument data using instrument vendor-provided algorithms and processing parameters
- L1b product performance and quality monitoring including image navigation and registration (INR)
- L1b product data delivery to the NESDIS Common Cloud Framework (NCCF)
- Radiometric performance trending and calibration of GeoXO instruments
- Spacecraft and instrument monitoring, performance trending, and parameter updates

- End-of-life de-commissioning and disposal of the GeoXO satellites to a super-synchronous orbit

Anomaly Investigation and Recovery (Time and Materials CLIN)

- Lead off-line investigation of satellite (spacecraft and instrument) and data anomalies with support from spacecraft, instrument, and Government personnel.
- Develop and execute Government-approved anomaly response plans and procedures

Special Engineering Analysis Tasks as requested by the Government (Time and Materials CLIN)

Estimated MIP Acquisition Timeline

- Draft RFP Release: Early 2027
- Final RFP Release: Summer 2027
- Contract Award: January 2028

RFI Response Instructions

The Government is seeking feedback from contractors with relevant mission capabilities and experience. Responses should include the vendor implementation approach and rough order magnitude costs of performing the functions listed below. Responses should be limited to no more than 30 pages.

1. Launch Vehicle Services
   a) Describe the vendor's capability and prior experience procuring launch vehicles for Government Class-B or similar satellite payloads.
   b) Describe the vendor's approach to procuring launch vehicles and satellite to launch vehicle campaign management from contract award through launch vehicle integration and launch for the GeoXO-1 and GeoXO-2 satellites.
   c) Provide a Rough Order Magnitude (ROM) cost for launch services for GeoXO-1 and GeoXO-2 broken out by satellite and fiscal year.
2) Payload Processing Services
   a) Describe the vendor's approach to providing a payload processing facility and supporting satellite launch site processing for GeoXO-1 and GeoXO-2.
   b) Describe the vendor's capability and prior experience in satellite launch site processing.
   c) Provide a ROM cost for payload processing services for GeoXO-1 and GeoXO-2 broken out by satellite and fiscal year.
3) Mission Operations Services
   a) Describe the vendor's overall approach to meeting the Government's GeoXO Operations Concept as defined above.
   b) Describe the vendor's approach to providing mission operations facilities, ground systems, antennas required to operate the GeoXO-1 and GeoXO-2 satellites.
   c) Describe the vendor's capabilities and prior experience performing on-orbit mission operations for a real-time critical mission including details on whether the ground systems were developed by your vendor or government furnished equipment (GFE).
   d) Describe the vendor's capabilities and prior experience maintaining and operating a FISMA High boundary ground system

e) Describe prior experience performing on-orbit commissioning and operations for optical imaging instruments
f) Describe prior experience with instrument calibration and navigated data generation and trending
g) Provide ROM costs for mission operations facilities, ground systems, and antennas to support the GeoXO-1 and GeoXO-2 satellites broken out by fiscal year.
h) Provide ROM costs for 10 years of on-orbit operations support for GeoXO-1 and GeoXO-2 broken out by satellite and fiscal year.

4) Vendor's overall assessment of the complexity, challenges, risks to performing elements 1, 2, and 3 above under a firm fixed price (FFP) contract

Additionally, the Government is seeking vendor-recommended alternatives to the Government MIP Operations Concept. MIP alternatives may include alternative contract structures and the use of GFE Government facilities and/or ground systems. Alternative MIP concepts should clearly describe the alternative approach, benefits, and the relative cost difference compared to the Government MIP Operations Concept.

5) Alternative MIP Operations Concepts - Vendors may recommend alternatives to the Government MIP CONOPS described above. For alternative approaches, the Vendor should provide the following:
   a) Describe the alternate approach
   b) State the benefits of the alternative approach
   c) Provide relative cost differences between the alternative approach and the Government MIP CONOPS
6) What, if anything, would you suggest changing regarding this acquisition?
7) What other insights would you like to provide?

Requirements/Specifications

1. Launch Vehicle Services
   a. Launch Planning Dates: December 2032 (GeoXO-1), October 2034 (GeoXO-2)
   b. Launch mass: 4595 kg
   c. Launch volumes: 5m Class Fairing, 4.529m Diameter x 7.303m High
   d. Direct to geostationary orbit (apogee and perigee altitudes 500km below GEO at 0 deg inclination with West CONUS separation)
2. Payload Processing Facility
   a. Occupancy timeline: 3-months from satellite arrival at PPF through launch
   b. Facility space: Accommodations for GeoXO Satellite processing (processing bay, hazardous propellant loading, launch vehicle encapsulation), satellite test and Electrical Ground Support Equipment (EGSE) control rooms, office space, and Ground Support Equipment (GSE) storage
   c. Cleanroom capability: Nominal ISO Class 8 environment with temporary ISO Class 7
   d. Accommodate spacecraft and instrument GSE
3. Mission Operations Services
   a. Spacecraft Communication Links

i. Command Uplink: S-band; 1kbps, 4 kbps, 64kbps; NIST FIPS 140-3 Encrypted
ii. Housekeeping Downlink Telemetry: L- and S-band; 1kbps, 4kbps, 100kbps
iii. Instrument Raw Data Downlink: X-band; 389 Mbps

Available Government Furnished Equipment (GFE) Resources:

The Government maintains the geostationary satellite operations resources listed below. RFI responses may include the use of these GFE resources. For any GFE resources incorporated in their response, the vendor should include any GeoXO implementation costs and the cost differences/savings as compared to the vendor providing all ground resources.

1. Wallops Command and Data Acquisition Station (WCDAS), Consolidated Back-Up (CBU) Command and Data Acquisition Station, and S-, L-, X-band capable antennas located in Wallops, VA and Fairmont, WV. The WCDAS and CBU antennas are capable of supporting GeoXO command and data acquisition without modification. WCDAS and CBU are multi-mission facilities that support NOAA GEO, LEO, and Space Weather missions.
2. NOAA Satellite Operations Facility (NSOF) located in Suitland, MD. The NSOF is a multi-mission satellite operations control center that supports all NOAA operational satellites.
3. NOAA Geostationary Earth Orbit (GEO) ground system consisting of mission management and L1b product generation systems. The NOAA GEO ground system is a multi-mission FISMA High boundary system that currently operates the GOES-R and Space Weather Follow-On missions. Development would be required to extend the GEO ground system to GeoXO missions.
4. NOAA NESDIS Common Cloud Framework (NCCF) for delivering GeoXO L0 and L1b product data and potential hosting of satellite command, control, and product generation systems.